

RUBTSOVSK, Russian Federation

WMO: 360340

Lat: 51.576N

Lon: 81.199E

Elev: 219

StdP: 98.72

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.5	(c) -32.0	(d) -38.3	(e) 0.1	(f) -35.0	(g) -35.0	(h) 0.1	(i) -31.9	(j) 14.0	(k) -6.5	(l) 12.9	(m) -7.0	(n) 2.0	(o) 210	(p) 0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	31.7	18.7	29.9	18.1	28.2	17.5	20.5	27.7	19.6	26.6	18.9	25.8	3.5	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.5	23.1	17.3	12.7	22.4	16.4	12.0	21.7	60.5	27.7	57.2	26.7	54.6	25.8	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	10.4	9.2	-37.2	35.1	5.0	1.8	-40.8	36.4	-43.7	37.4	-46.5	38.5	-50.1	39.8
			-37.3	22.2	4.9	1.3	-40.8	23.1	-43.7	23.9	-46.5	24.6	-50.1	25.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-16.8	-14.5	-6.1	5.7	13.6	19.4	20.7	18.6	12.2	4.9	-4.9	-12.8
	DBStd	14.60	8.37	7.86	6.93	5.65	4.91	3.87	3.20	3.68	4.50	5.18	7.75	7.78
	HDD10.0	3538	830	686	499	149	23	1	0	1	28	170	447	706
	HDD18.3	5662	1088	919	757	379	160	35	14	43	189	416	697	964
	CDD10.0	1138	0	0	0	20	133	282	331	266	94	13	0	0
	CDD18.3	221	0	0	0	0	12	67	87	50	5	0	0	0
	CDH23.3	2329	0	0	0	9	191	684	777	564	103	1	0	0
	CDH26.7	734	0	0	0	1	50	215	258	183	28	0	0	0
(14) Wind	WSAvg	4.0	4.2	4.2	4.2	4.2	4.2	3.6	3.1	3.1	3.3	4.1	4.6	4.8
(15) Precipitation	PrecAvg	341	18	19	20	20	32	32	57	36	28	31	27	20
	PrecMax	570	80	58	70	58	74	73	150	81	70	77	59	45
	PrecMin	176	1	2	7	1	1	2	8	8	2	7	2	2
	PrecStd	78	16	14	16	13	20	23	40	20	16	16	13	11
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.5	2.2	12.1	24.5	31.6	32.8	34.2	33.8	30.1	22.1	12.3	2.1
		MCWB	0.1	0.2	6.3	13.2	17.5	18.3	20.3	20.1	16.0	11.4	6.5	0.3
	2%	DB	-1.1	0.2	7.2	20.9	27.7	30.9	31.4	30.8	26.2	18.4	9.3	0.7
		MCWB	-2.5	-1.3	3.8	11.4	15.2	18.0	19.6	18.4	14.8	10.0	4.8	-0.7
	5%	DB	-3.0	-1.5	4.6	18.2	25.4	29.1	29.5	28.7	23.5	15.6	6.7	-0.8
		MCWB	-4.2	-2.7	2.2	10.0	14.3	17.4	18.9	17.9	14.0	8.9	3.6	-2.1
	10%	DB	-4.9	-3.2	2.7	15.5	22.9	27.4	27.6	26.6	20.7	12.7	4.2	-2.5
		MCWB	-5.9	-4.5	0.9	8.5	13.3	17.1	18.5	17.1	12.7	7.5	2.1	-3.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	0.7	6.6	13.8	18.3	20.3	22.5	21.4	17.2	12.2	7.6	0.9
		MCDB	1.3	1.7	11.0	22.1	29.3	28.5	29.3	30.1	26.3	18.7	10.1	1.7
	2%	WB	-2.2	-1.2	4.1	12.0	16.5	19.4	21.1	19.8	15.7	10.7	5.2	-0.7
		MCDB	-1.3	-0.1	6.9	19.4	24.7	27.3	27.9	27.7	24.0	17.3	8.0	0.4
	5%	WB	-4.1	-2.7	2.3	10.5	15.2	18.7	20.1	18.8	14.6	9.4	3.6	-2.1
		MCDB	-3.0	-1.4	4.3	17.1	23.1	26.3	26.4	26.4	22.3	14.7	6.8	-0.9
	10%	WB	-6.0	-4.4	1.0	9.0	14.1	17.9	19.4	17.7	13.4	8.0	2.1	-3.5
		MCDB	-4.9	-3.2	2.5	15.2	21.5	25.3	25.6	24.9	19.7	12.4	4.2	-2.4
(35) Mean Daily Temperature Range	MDBR		8.7	10.0	9.4	11.5	12.5	11.7	10.9	12.2	12.1	9.6	7.5	8.3
	5% DB	MCDBR	8.5	9.3	9.8	16.3	16.4	14.8	14.1	15.9	17.1	14.9	9.5	8.2
		MCWBR	8.0	8.4	7.0	8.4	7.1	5.3	4.9	5.9	7.6	8.0	6.4	7.7
	5% WB	MCDBR	8.5	9.5	9.0	15.2	13.8	12.6	11.5	13.4	15.3	13.6	8.5	8.5
		MCWBR	8.2	8.9	6.6	8.3	6.8	5.1	4.7	5.7	7.3	7.8	6.0	8.2
(40) Clear-Sky Solar Irradiance	taub		0.300	0.338	0.389	0.413	0.415	0.413	0.434	0.410	0.370	0.356	0.342	0.308
	taud		2.029	1.950	1.924	2.127	2.177	2.270	2.214	2.253	2.310	2.239	2.081	1.981
	Ebn at Noon		663	740	778	820	838	843	816	816	809	731	603	566
	Edh at Noon		83	120	151	139	139	128	133	122	102	90	79	74
(44) All-Sky Solar Radiation	RadAvg		1.33	2.46	3.91	4.72	6.04	6.43	5.82	5.15	3.81	2.08	1.09	0.92
	RadStd		0.10	0.11	0.33	0.36	0.43	0.51	0.39	0.28	0.30	0.26	0.12	0.12

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	-0.52	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page